

Topic 10: Testing & Verification — Pytest Suite & E2E Validation

TOPIC OBJECTIVE

In clinical decision support software, untested edge cases represent direct patient malpractice risks. AegisClinical enforces an exhaustive 4-tier testing pyramid in tests—spanning sub-millisecond mathematical rule boundaries, multi-resource FHIR parser ingestion, parallel LangGraph state reducers, and FastAPI HTTP DoS security constraints—achieving 100% test pass rates across all clinical scenarios.

1. The Real-World Analogy: The FAA Flight Simulator & Wind Tunnel

An aerospace engineering team would never evaluate a dual-engine flameout or a hydraulic failure while cruising with 300 passengers on a commercial airliner. They place the aircraft in high-fidelity wind tunnels, hydraulic structural stress rigs, and computerized flight simulators.

In clinical AI software, an engineering team must never discover that a creatinine delta threshold evaluates incorrectly (e.g. 0.29 vs. 0.30 mg/dL) while a living patient is deteriorating in the ICU. Automated pytest suites represent the **clinical pre-flight stress simulator**: they rigorously test boundary conditions, malformed payloads, and extreme drug combinations before code touches a clinical ward.

2. The AegisClinical 4-Tier Testing Pyramid

Testing Tier	Target Module & Suite	Clinical Verification Focus
Tier 1: Rules & Math Sub-millisecond	tests/test_rules.py (Deterministic Rules)	KDIGO Stage 1/2/3 mathematical cutoffs (≥ 0.3 mg/dL, 2.0x, 3.0x), panic electrolyte thresholds ($K^+ \geq 6.0$), and drug name normalization.
Tier 2: Data Ingestion FHIR R4 Validation	tests/test_parser.py (Synthea Parser)	Multi-hundred resource FHIR bundle traversal, chronological lab trajectory sorting, and medication/condition code mapping.
Tier 3: Multi-Agent E2E Graph Integration	tests/test_graph.py (LangGraph State)	Scatter-gather agent execution, operator.add reducer concatenation, and final coordinator triage assessment synthesis.
Tier 4: API & Security HTTP Contract	tests/test_api.py (FastAPI TestClient)	CORS headers, health check heartbeat, patient list routes, and 5MB payload DoS rejection (HTTP 413).

3. Concrete Test Implementation: Boundary Conditions & Interactions

Below is the exact pytest assertion from tests/test_rules.py and tests/test_graph.py verifying that borderline KDIGO transitions and life-threatening Triple Whammy combinations trigger flawlessly:

```
def test_kdigo_aki_stages():
    # Baseline 1.0 -> 1.0 (Stable): No alert
    assert evaluate_kdigo_aki(1.0, 1.0) is None
    # Stage 1: Absolute surge delta >= 0.3 mg/dL
    res_stage1 = evaluate_kdigo_aki(1.0, 1.35)
    assert res_stage1[0] == "KDIGO_AKI_STAGE_1"
    # Stage 2: Surge >= 2.0x baseline (1.0 -> 2.4 mg/dL)
    res_stage2 = evaluate_kdigo_aki(1.0, 2.4)
    assert res_stage2[0] == "KDIGO_AKI_STAGE_2"
    assert res_stage2[1] == "HIGH"

def test_drug_interactions_triple_whammy():
    meds = ["Lisinopril 20 MG", "Furosemide 40 MG", "Ibuprofen 600 MG"]
    interactions = evaluate_drug_interactions(meds)
    assert len(interactions) == 1
    assert interactions[0].severity == "CONTRAINDICATED"
    assert "Triple Whammy" in interactions[0].mechanism
```

4. Full-Suite Pytest Execution Matrix

Test Suite File	Items	Duration	Execution Status
tests/test_rules.py	6 test cases	< 0.05s	PASSED (100%)
tests/test_parser.py	3 test cases	< 0.12s	PASSED (100%)
tests/test_graph.py	3 test cases	~ 9.20s	PASSED (100%)
tests/test_api.py	5 test cases	< 0.45s	PASSED (100%)

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- **Topic 4:** The Ingestion Parser — Extracting Trajectories from FHIR Bundles
- **Topic 5:** Deterministic Rules Engine — Mathematical KDIGO & DDI Matrices
- **Topic 6:** LangGraph & State Contracts — Parallel Reducers with operator.add
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